Paper Winder

Owner's Manual

Sandkey Services paper winder (M1)

```
-= PAPER WINDER =-
[1] 30ft [2] 90ft
```

Covers firmware version **fw=6** and the PC program **winder**

Manual revision: **2026-10-21**

Keep this manual near the machine. Read Chapter 2 (Safety) before you use the machine for the first time. The quick-reference card in Appendix A can be cut out and posted beside the panel.

Support: [SUPPORT CONTACT] Machine serial number: [SERIAL NUMBER]

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Chapter 1

Quickstart

Wind one roll in eight steps

1. **Power off. Load and thread the paper** past the measuring wheel the way you were shown at installation. Keep hands clear of the roll and the wheel.
2. **Power on and wait one second.** The screen shows `-= PAPER WINDER =-` and `[1] 30ft [2] 90ft`. **LED 1** and **LED 3** are lit.
3. **Push Button 1 for 30 feet, or Button 2 for 90 feet.** One push starts the wind. The motor runs and the screen shows `WINDING...`, a percent bar and the length.
4. **Let it finish.** About 6 feet before the end the motor stops, then the screen shows `TRIMMING...` and the motor makes bumps that get shorter and shorter to finish the length (this can take up to about half a minute). The wind stops about 2 inches *short* of the label on purpose.
5. **Two long beeps and `** PULL SLACK **`** mean the wind is complete but stopped about 2 inches short on purpose. The second row says how much, for example `PULL 2.0 in BY HAND`. **Pull that much paper out by hand** (gently, towards the roll) to take up the slack, then cut the paper and remove the roll.
6. **Three short beeps and `DONE`** mean the roll is complete with nothing to pull: the full length went past the wheel. Cut the paper and remove the roll.
7. **Long beeps and `!! UNDER TARGET !!`** mean the roll is *not* guaranteed. Push any button to silence the alarm, then see Section 5.8.
8. **To stop at any time, push Button 1 or Button 2.** In an emergency switch the machine's power off.

Other buttons. Button 3 pauses and resumes a wind. **Button 4** turns the motor on and off by hand ("manual") from the idle screen, or nudges the paper while a wind is paused.

One push = one action. The panel buttons are latching switches: a push leaves the button sitting in, the next push lets it pop out. The machine counts *every* push, in or out, as one click. Which way a button is resting does not matter. After a wind it is normal for a button to stay pushed in.

You do not need the PC to wind paper. The PC program is only for calibration and settings (Chapter 10).

Chapter 2

Safety

Read this page before you use the machine. Post the quick-reference card (Appendix A) beside the panel.

WARNING – MOVING PARTS

The roll, the motor drive and the measuring wheel can pull in fingers, hair, loose sleeves, gloves and jewellery. Switch the machine's power off before you load paper, thread paper, clear a jam, clean the wheel or reach into the paper path.

WARNING – THE MOTOR CAN START WITHOUT WARNING

The motor starts the moment someone pushes **Button 1**, **Button 2** or **Button 4**, or when a person at a connected PC uses the program's Test mode. During a wind the motor stops and starts again by itself (the trim bumps, Section 5.3). A wind is not over until you hear the beeps and see the result screen. Always assume the motor can move while power is on.

WARNING – THE BUTTONS ARE NOT AN EMERGENCY STOP

Button 1 and **Button 2** stop a wind through the controller. If the controller has a fault, the screen is frozen, or the motor does not stop, **switch the machine's power off**. The paper keeps moving for a short distance after the motor stops (the "coast", up to a few yards at full speed).

General rules

- Only trained staff should run the machine. Keep children and visitors away.
- Manual mode (**Button 4**) runs the motor continuously until you push **Button 4** again. Never

leave the machine in manual mode unattended.

- The machine does *not* detect a jam during the main part of the wind (only during the trim). If the paper stops, bunches or tears, push **Button 1** or **Button 2** at once, or switch the power off.
- If someone is using the PC program on the machine, agree who is in charge. While Test mode is on, the panel cannot start a wind, but **Button 1** and **Button 2** still stop the motor.
- Do not open the control box with the power on. Only a qualified person should work on the wiring, the relay or the motor.
- Connect only the three wires of the USB serial adapter (TX, RX, GND) described in Section 10.1. Do not power the machine from the adapter.
- Keep the work area dry and the cables out of walkways.
- Machine power rating and supply: [POWER RATING – SEE MACHINE LABEL].

If something goes wrong: switch the power off, wait until all motion has stopped, then see Chapter 14.

Chapter 3

Machine overview

The paper winder pulls paper off a supply, past a rubber measuring wheel, and onto a roll. A small controller board counts how far the measuring wheel turns, switches the motor on and off through a relay, and shows what it is doing on a four-line screen. You start and stop winds with four buttons on the panel.

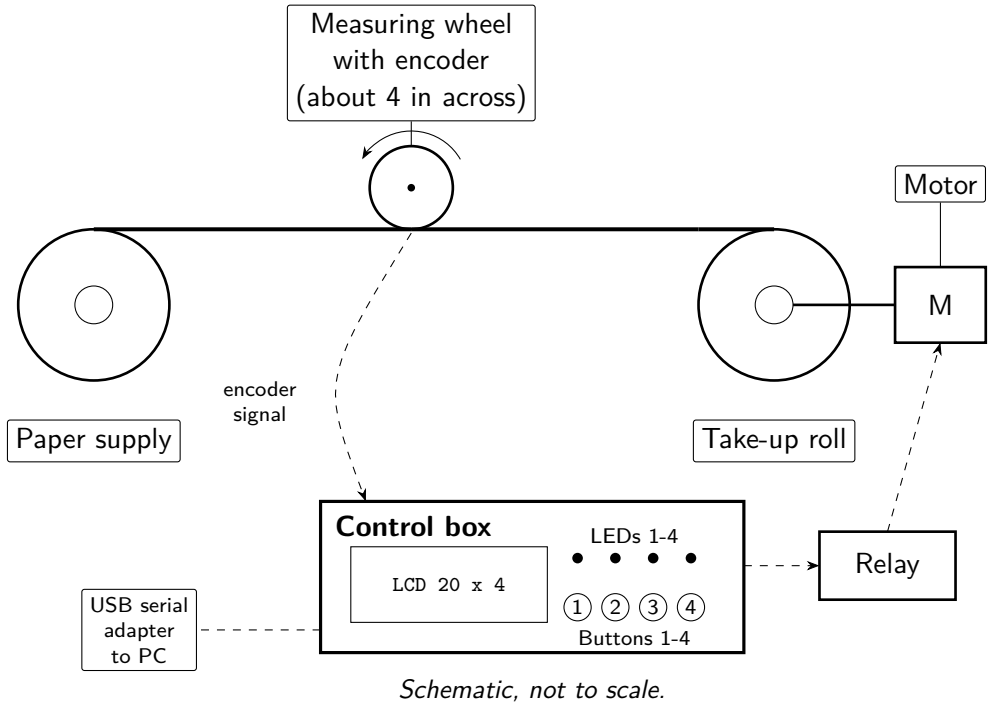


Figure 3.1: The main parts of the paper winder.

The main parts

Measuring wheel and encoder
A rubber-faced wheel about 4 inches across rides on the paper. A rotation sensor (the encoder) on its shaft gives 2400 counts for every turn of the wheel, which is about 191 counts per inch of paper. This is the only thing the machine uses to measure length, so the wheel must be clean and must turn freely with the paper without slipping.

Motor and relay
The controller switches the motor on and off through a relay. It is simply on or off; there is no speed control. When the relay opens, the roll keeps turning for a short distance (the *coast*).

Control box
Holds the controller board, the 20-character by 4-line screen, the four LEDs, the four buttons and a buzzer.

Serial connection
Three wires (TX, RX, GND) from a USB serial adapter let a PC run the **winder** program for calibration and settings. The PC is not needed to wind paper.

USB port on the controller
Used only to load new firmware (Chapter 12). It does not talk to the winder program.

NOTE

Your machine's exact layout (which side the supply is on, where the power switch is) may differ from the drawing. The power switch location is: [POWER SWITCH LOCATION].

Chapter 4

Panel buttons and LEDs

4.1 How the buttons work

The four panel buttons are **latching** switches. Push one and it stays pushed in; push it again and it pops back out. The machine does not care which way a button is resting. **Every push, in or out, counts as one click**, and one click does one thing.

- After a wind a button may still be sitting pushed in. That is normal. You do not need to pop it out before the next wind.
- A button that is already pushed in when you switch the power on does nothing. The machine ignores button positions at power-up. Wait about one second after power-up before you push a button.
- Very fast double pushes may count as one: the machine ignores contact chatter shorter than about 1/60 of a second.
- While the PC program has the machine in Test mode, the panel cannot start anything (Section 10.7).

4.2 Button map

Button	At the idle screen	During a wind
Button 1	Start a **30 foot** wind.	**Stop** the wind.
Button 2	Start a **90 foot** wind.	**Stop** the wind.
Button 3	Nothing.	**Pause**; push again to **resume**.
Button 4	**Manual**: motor on; push again for motor off.	While *paused*: motor on/off to nudge the paper by hand (stops by itself at the target length). While winding: nothing.
Any button	Clears an `UNDER TARGET` alarm *and* does its normal job.	

Tip

To stop quickly, push *either* **Button 1** or **Button 2**; it does not matter which one started the wind.

4.3 LEDs

The four LEDs above the buttons tell you the state at a glance.

LEDs lit	Meaning
LED 1 and **LED 3**	Idle, ready to wind.
LED 4 only	Manual mode: the motor is on.
None, then 1, 1–2, 1–3, all four	Winding: a bar graph that fills in quarters as the wind goes on.
1–3, then all four	Trimming: all four light once the length has passed the target.
LED 2 and **LED 4**	Test mode (PC program has control).
Fault LED (where fitted)	The controller found an internal fault. Switch off, then on; if it comes back, call [SUPPORT CONTACT].

4.4 Buzzer

The buzzer beeps when a wind finishes (two long beeps if you must pull the slack), and sounds an alarm when a roll comes up short. See Chapter 8 for every pattern.

Chapter 5

Winding paper step by step

5.1 Before you start

- Power off. Load the paper supply and thread the paper under the measuring wheel and onto the take-up core, as you were shown at installation ([THREADING DIAGRAM – SEE INSTALLER NOTES]).
- Check that the measuring wheel sits on the paper and turns freely.
- Power on. Wait about one second. The idle screen appears and **LED 1** and **LED 3** light.

```
-= PAPER WINDER =-

[1] 30ft    [2] 90ft
[4] click = manual
```

5.2 Start a wind

Push **Button 1** for a **30 foot** roll or **Button 2** for a **90 foot** roll. One push is enough. Starting a wind also silences any buzzer that was still sounding.

The relay closes, the motor runs and the screen shows the progress:

```
WINDING...
 45% [#####--------]
LEN   12.3/  30.0 ft
[1/2]STOP   [3]PAUSE
```

The LEDs fill up in quarters as the wind goes on. If the machine has seen encoder errors, the top row shows `WINDING ERR:n` instead (see Section 7.4).

5.3 The end of a wind: coast and trim

The motor cannot stop the roll instantly. After the relay opens the roll keeps turning for a short distance, called the **coast** (set at the factory to 8 feet, more than the roll is expected to glide). So the machine switches the motor off about one coast *before* the goal, and lets the roll glide towards it.

Then it **trims**:

1. The screen shows `TRIMMING...`. The machine waits half a second for the paper to stop.
2. If the length is not yet enough, it gives the motor a bump and waits again. It measures how far the roll glided after the motor stopped and sizes each bump so that the glide lands at the goal, so the bumps get shorter as the gap closes and the last ones are tiny. It repeats this until the length is at least the goal: the target plus the safety margin (0 from the factory), less the 2 inch slack (see Section 5.4). This can take up to about half a minute. What it measures is used for this roll only; turning the roll by hand while the machine is idle changes nothing.
3. When the length is enough, it beeps and shows the result: **two long beeps** and `** PULL SLACK **` if the roll is a little short of the label on purpose (the usual case), or **three short beeps** and `DONE` if the roll reached the label length.

```
TRIMMING...
 99% [###########-]
LEN   29.9/  30.0 ft
[1/2]STOP   [3]PAUSE
```

5.4 Stopping short on purpose: pull the slack

The machine aims for the target length *minus 2 inches* (the **slack**). It does this so that it never over-winds, and so that the paper is slightly slack between the wheel and the roll for you to tidy up. When the wind ends 2 inches or so short of the label, the machine makes it obvious:

- two **long beeps** (0.7 s each), once;
- the screen shows `** PULL SLACK **` and, on the next row, `PULL 2.0 in BY HAND` (the figure is the actual shortfall).

```
** PULL SLACK **
PULL 2.0 in BY HAND
LEN 29.9/30.0 ft
[1] 30ft    [2] 90ft
```

Pull that much paper out by hand (gently, so the wheel is not lifted) to take up the slack, then cut the paper and remove the roll. This is a good result and the usual one. If the roll happens to end on or past the label length, there is nothing to pull and you get `DONE` with three short beeps instead.

5.5 Finished

```
-= PAPER WINDER =-
DONE 30.1/30.0 ft
+4.3 in over
[1] 30ft    [2] 90ft
```

The second row shows the measured length and the target. The third row says how far over (or short) the roll is compared with the target, in inches (or in feet when it is a foot or more). Just after the motor stops the screen shows `NOW` and `settling...` for a moment while the roll comes to rest; then `DONE` and the final figure. Cut the paper and remove the roll. The result stays on the screen for 60 seconds, then the normal idle screen returns. You can start the next wind straight away; a button still sitting pushed in is fine.

5.6 Pause, nudge and resume

- Push **Button 3** to **pause**. The motor stops and the screen shows `PAUSED`. The bottom row changes to `[3]PLAY`.
- While paused, push **Button 4** to run the motor by hand (the screen shows `PAUSED - MANUAL RUN`); push **Button 4** again to stop. The motor also stops by itself if the length reaches the target.
- Push **Button 3** again to **resume**. The wind carries on from where it was; the hand-run length counts towards the roll.

5.7 Stop a wind early

Push **Button 1** or **Button 2**. The motor stops at once. What you see next:

- If the roll had already reached the goal, it is reported as usual: `** PULL SLACK **` or `DONE`.
- Otherwise the screen shows `!! UNDER TARGET !!` with `STOPPED` and how much is missing. There is no alarm sound for a stop you made yourself. The screen stays until you push a button.

5.8 When the roll comes up short

If the trim cannot finish (the paper does not move when bumped, too many bumps, or the trim takes longer than 45 seconds), the machine gives up and sounds the **under-target alarm** (not to be confused with the two beeps of the slack notice): three long beeps, repeated every 6 seconds for about a minute.

```
!! UNDER TARGET !!
SHORT -8.2 in
LEN 29.8/30.0 ft
[1] 30ft    [2] 90ft
```

What to do:

1. Push any button to silence the alarm. (That push also does its normal job, so **Button 1** would start a new 30 foot wind. Push **Button 3** if you only want to clear the screen.)

2. Switch the power off and look for the cause: paper run out, a jam, the wheel lifted off the paper, a torn web.
3. Do **not** sell the short roll as full length. Either rewind it or add the missing length by hand.
4. See Chapter 14 if it keeps happening.

5.9 Manual mode

From the idle screen, push **Button 4**: the motor runs continuously and the screen shows `MANUAL: motor ON`; only **LED 4** is lit. Push **Button 4** again to stop. Use this for threading and for winding odd lengths by eye. The machine does not count or stop a manual run for you.

WARNING

Never leave the machine in manual mode unattended.

5.10 After a power cut or reset

If the power fails, or the controller resets itself (watchdog), the relay opens, the motor stops, and the machine starts up at the idle screen. Any wind in progress and its result are lost. Measure the roll by hand or rewind it.

Chapter 6

The minimum-amount guarantee

The machine is built so that a customer never gets less paper than the label says. Every wind ends in one of three ways:

Result	What it promises
`** PULL SLACK **`, two long beeps	The roll is within the slack (2 inches) of the full length **plus the margin** (margin 0 from the factory). It is short only by the amount shown; pull that much paper out by hand and the roll is full length.
`DONE`, three short beeps	At least the full length plus the margin went past the measuring wheel. The roll is good.
`!! UNDER TARGET !!`	The machine could **not** confirm the full length. Treat the roll as short. Alarm beeps unless you stopped the wind yourself.

6.1 How it works

1. The **goal** is the target length plus the margin, less the **slack** of 2 inches. With the factory margin of 0, a 30 foot roll has a goal of 30 feet less 2 inches.
2. The motor runs until the paper is one *coast* (8 feet from the factory) short of the goal, then it is switched off and the roll glides on. The coast is set generously on purpose: running short is fixed by the trim, but paper that has gone past the goal cannot be taken back.
3. After the paper has stopped, the machine compares the count with the goal. If it is short, it bumps the motor briefly and checks again. It watches how far the roll glides after each stop

and sizes the next bump so that the glide it expects lands at the goal; the last bumps are tiny. What it learns is used for that roll only and forgotten when the next wind starts.

4. It only reports a good roll when the stopped count is at or past the goal: `** PULL SLACK **` if that is still short of the label length (pull the slack out by hand), `DONE` if the label length was reached. It never guesses: the decision is made on paper that has stopped moving.

6.2 What the guarantee depends on

The guarantee is only as good as the measurement. It assumes:

- The measuring wheel is **calibrated** (Chapter 11) and its rubber is clean and not worn.
- The wheel turns with the paper **without slipping** and is not lifted off the paper.
- The paper goes onto the roll; a tear or a jam upstream of the wheel will not be noticed while the motor is running (only during the trim).

6.3 How much over?

A normal roll is within about 2 inches of the label, usually a little short (the slack you pull out by hand) or a little over from the glide after the last bump. When the real coast of your machine is under about 5 feet the design estimate is well under an inch past the goal on average and no more than about 2 inches past the goal; a roll that finishes with bumps is at most about 3 inches past the goal. If the real coast is longer than the 6 foot setting, the main run itself goes past the goal by the difference, and the trim cannot take that back. These figures are estimates from the design and have not yet been measured on the bench (Chapter 15). If your rolls are regularly much longer than that, recalibrate the coast (Chapter 11).

Tip

The `DONE` screen tells you the exact overrun of every roll, for example `+4.3 in over`, and `PULL SLACK` the exact shortfall. Glance at it each time: a sudden change is the first sign of a dirty or slipping wheel.

Chapter 7

Screen catalogue

The screen has four rows of 20 characters. This chapter shows every screen the machine can display, with what it means and what you can do. Numbers on the screens below are examples.

If the screen goes blank or shows garbage

Electrical noise from the motor relay can upset the screen. The machine redraws it after every motor start and stop and every 2 seconds anyway, so a garbled screen should fix itself within about 2 seconds. If it does not, see Chapter 14.

7.1 Idle

```
-= PAPER WINDER =-

[1] 30ft    [2] 90ft
[4] click = manual
```

Ready. **Button 1** starts 30 feet, **Button 2** starts 90 feet, **Button 4** turns the motor on by hand. **LED 1** and **LED 3** are lit.

7.2 Manual mode

```
-= PAPER WINDER =-
MANUAL: motor ON
[1] 30ft    [2] 90ft
[4] click = auto
```

The motor is running because **Button 4** was pushed. Push **Button 4** again to stop it (the bottom row reminds you that this returns to automatic). Only **LED 4** is lit. **Button 1** and **Button 2** still start a wind from here.

7.3 Encoder miss warning

```
-= PAPER WINDER =-
!! ENC MISS:3      !
[1] 30ft    [2] 90ft
[4] click = manual
```

The controller saw signals from the measuring-wheel encoder that it could not make sense of (the number is how many). The length count may be slightly wrong. Check the encoder cable and connector, then see Chapter 14. The same line can appear on the bottom row of a result screen.

7.4 Winding

```
WINDING...
 45% [#####--------]
LEN   12.3/  30.0 ft
[1/2]STOP   [3]PAUSE
```

Row 2: percent done and a bar. Row 3: length so far and the target, in feet (and tenths of a foot). **Button 1** or **Button 2** stops, **Button 3** pauses.

If the machine has counted encoder errors during the wind, the top row shows them:

```
WINDING    ERR:2
 45% [#####--------]
LEN   12.3/  30.0 ft
[1/2]STOP   [3]PAUSE
```

7.5 Paused

```
PAUSED
 45% [#####--------]
LEN   12.3/  30.0 ft
[1/2]STOP   [3]PLAY
```

The motor is stopped. **Button 3** resumes, **Button 4** runs the motor by hand, **Button 1** or **Button 2** ends the wind.

```
PAUSED - MANUAL RUN
 52% [######-------]
LEN   15.6/  30.0 ft
[1/2]STOP   [3]PLAY
```

Paused, with the motor running by hand (**Button 4**). Push **Button 4** to stop the motor, or **Button 3** to resume the automatic wind.

7.6 Trimming

```
TRIMMING...
 99% [############-]
LEN   29.9/  30.0 ft
[1/2]STOP   [3]PAUSE
```

The main run is over. The motor is bumped briefly until the length is reached. This normally takes a few seconds.

7.7 Result: settling and done

```
-= PAPER WINDER =-
NOW  30.0/30.0 ft
settling...
[1] 30ft    [2] 90ft
```

Shown for a moment while the roll comes to rest.

```
-= PAPER WINDER =-
DONE 30.1/30.0 ft
+4.3 in over
[1] 30ft    [2] 90ft
```

The roll is complete. Row 3 can read `+4.3 in over`, `on target`, or, for a large overrun, `+1.3 ft over`. A miss under one foot is shown in inches, from one foot up in feet. After a legacy-mode wind (Section 9.2) it can also read a negative figure such as `-1.3 in short`. The screen returns to idle by itself after 60 seconds. You can start the next wind at once.

7.8 Pull the slack

```
** PULL SLACK **
PULL 2.0 in BY HAND
LEN 29.9/30.0 ft
[1] 30ft    [2] 90ft
```

The wind is complete, but the machine stopped a little short of the label on purpose (factory: about 2 inches). Two long beeps sound once. Row 2 says how much paper to pull by hand; row 3 shows the length counted and the target. **Pull that much paper out by hand** to take up the slack, then cut and remove the roll. This is a good result, not a fault. Push any button to silence the beeps; the

screen returns to idle by itself after 60 seconds. If the roll stopped on or past the label length you see the plain DONE screen instead.

7.9 Under target

```
!! UNDER TARGET !!
SHORT -8.2 in
LEN 29.8/30.0 ft
[1] 30ft    [2] 90ft
```

The trim could not reach the goal (target plus margin, less the 2 inch slack). Row 2 shows how much is missing, measured against the goal. The alarm sounds. The screen stays until you push any button; that push also does its normal job.

```
!! UNDER TARGET !!
STOPPED -22.6 in
LEN 29.4/30.0 ft
[1] 30ft    [2] 90ft
```

You stopped the wind with **Button 1** or **Button 2** before it reached the goal. No alarm sounds. The roll is short by the amount shown.

7.10 Test mode (PC in control)

```
-= PAPER WINDER =-
DEBUG: HOST CONTROL
panel buttons locked
[1/2] = STOP motor
```

A PC running the **winder** program has put the machine in Test mode (Section 10.7). The panel will not start a wind. **Button 1** or **Button 2** stops the motor and ends Test mode. **LED 2** and **LED 4** are lit.

Chapter 8

Beeps and alarms

The machine has one buzzer and only a few sounds. Learn them; they tell you the result of a wind without looking.

Sound	When	What to do
Two long beeps, once (beep 0.7 s, gap 0.25 s, twice)	Wind finished, `** PULL SLACK **`.	Good roll, stopped about 2 inches short on purpose. Pull the slack out by hand (the screen says how much), then cut and remove it.
Three short beeps (beep 0.3 s, gap 0.15 s, three times)	Wind finished, `DONE`.	The roll reached the label length. Cut and remove it.
Three long beeps, repeated (beep 0.7 s, gap 0.25 s, three times; repeats every 6 s, up to 10 times, about one minute)	`!! UNDER TARGET !!` with `SHORT`.	The roll is short. Push any button to silence. Find the cause (Section 5.8).
Silence	You stopped a wind with **Button 1** or **Button 2**.	Read the screen: `DONE`, `PULL SLACK` or `STOPPED`.
Buzzer stops suddenly	A new wind was started.	Normal: starting a wind always silences the buzzer.
Three short beeps on finish *and* on stop	Only in legacy mode (bump setting 0, Section 9.2).	No guarantee in this mode; check every roll.

Notes

- The two beeps of the slack notice are as long as the alarm beeps but there are only two and they do not repeat. Look at the screen if you are not sure which you heard.
- There is no beep at power-up and no beep when you push a button.
- The alarm stops by itself after about one minute, but the `UNDER TARGET` screen stays until someone pushes a button. If you come back to a silent machine showing `UNDER TARGET`, the last roll is short.
- A power cut or a controller reset silences everything and loses the result.

Chapter 9

Settings

The machine keeps seven numbers in its own memory. They survive power-off. You change them only from a PC with the **winder** program (Chapter 10); there is no settings menu on the panel. Most owners never need to touch them: calibration (Chapter 11) sets the important ones for you.

The machine counts length in *encoder counts*. From the factory, one inch of paper is about 191 counts, and one count is about 0.005 inch. The program converts to inches and feet for you.

Setting (name in the program / short name)	Factory value	What it does, allowed range
Measuring wheel size `wheel_um`	4.0462 in across (102773 micrometres)	Wheel diameter. Turns counts into inches. Range 50000 to 300000 micrometres (about 2 to 11.8 in).
Encoder counts per wheel turn `cpr`	2400	Counts per full turn of the wheel. Must match the encoder (600 lines x 4). Range 100 to 100000.
Short wind (30 ft button) `short`	30.0 ft (67970 counts)	Target length for **Button 1**. Range 1000 to 4000000 counts.
Long wind (90 ft button) `long`	90.0 ft (203911 counts)	Target length for **Button 2**. Same range.
Coast after the motor stops `coast`	96 in (18125 counts)	How far the roll glides after the motor is switched off. The motor is cut this far before the goal. Must be smaller than both targets. Set by calibration. Saved settings keep their coast after a firmware update: use **Reset to factory settings** and Save to get the 8 ft default.
(not in the program) `margin`	0 (0 counts)	Extra length added to every roll for the guarantee. The wind aims 378 counts (2.0 in) short of target plus margin, so a margin of 378 or more cancels the slack. Range 0 to 2000 counts (about 0 to 10.5 in).
(not in the program) `bump`	1 (on)	1 = trim mode with the minimum-amount guarantee. 0 = legacy mode (no trim, no guarantee).

One more rule: the longer target plus the margin may not exceed 4000000 counts (about 1700 feet). The machine refuses settings that break any rule.

9.1 Changing a setting

1. Start the **winder** program and open the **Settings** tab.
2. Type the new value. A banner **Unsaved changes** appears.
3. Click **Save to the machine**. Nothing changes on the machine until you do. **Undo all** throws your edits away.

Reset to factory settings puts every value back to the table above. Check the screen afterwards: if the **Unsaved changes** banner shows, click **Save to the machine** to store them.

Change the wheel and counts only with care

A wrong **Measuring wheel size** or **Encoder counts per wheel turn** makes every roll the wrong length. After changing either, run a calibration with the length check (Chapter 11).

Saved settings from older firmware

A machine that already has saved settings keeps its old 30 yd and 90 yd lengths, its old coast and margin after the firmware is updated. To get the 30 ft and 90 ft buttons and margin 0, connect the PC program, click **Reset to factory settings** and then **Save to the machine**.

9.2 Margin and legacy mode

The `margin` and `bump` settings are not shown in the program and calibration never changes them. They can only be changed through the serial command line by a technician (see `docs/hardware/serial-protocol.md` in the source repository). Leave `bump` at 1. With `bump` at 0 the machine behaves like the older firmware: it cuts the motor once and accepts wherever the roll stops, with no trim, no `UNDER TARGET` check and no guarantee.

9.3 When settings are lost

The settings are stored with a check code. If the stored settings are damaged, or were saved by firmware older than fw=6, the machine ignores them and uses the factory values. After updating to fw=6 from an older version you must calibrate (or re-enter your numbers) and save once. The program's **Connection check** reports this as **Saved settings are valid** failing.

Chapter 10

The PC program (winder)

winder is a small program for a Windows, Mac, Linux or Chromebook computer. It shows the machine's state, calibrates it and edits the settings. It runs a page in your web browser; nothing is sent over the internet. You do *not* need it to wind paper.

10.1 What you need

- A computer.
- A **USB serial adapter** with 5 volt logic (common chips: CH340, CP210x, FTDI). Most computers have the driver built in.
- Three wires from the adapter to the controller board:

Adapter pin	Controller pin
TX	D0 (RX)
RX	D1 (TX)
GND	GND

WARNING

Do not connect the adapter's 5 V or 3.3 V power pin. The machine has its own power. The controller's own USB port is *not* used by the winder program (it is only for loading firmware, Chapter 12).

10.2 Download

Get the program from the release page `https://github.com/lognd/stpone/releases/tag/winder-latest`. Pick the file for your computer:

Computer	File
Windows	`winder-windows-x86_64.exe`
Mac with Intel chip	`winder-macos-x86_64.tar.gz`
Mac with Apple chip (M1 and later)	`winder-macos-aarch64.tar.gz`
Linux PC	`winder-linux-x86_64.tar.gz`
Linux on ARM, Raspberry Pi	`winder-linux-aarch64.tar.gz`
Chromebook	see Section 10.4

The release is rebuilt automatically whenever the software passes all its tests, so "latest" is always the current version.

10.3 Start the program

Windows

Double-click `winder-windows-x86_64.exe`. If Windows warns about an unknown publisher, choose to run it anyway.

Mac and Linux

Open a terminal in the download folder and type (for example)

```
tar -xzf winder-linux-x86_64.tar.gz
./winder
```

The program opens your web browser at `http://127.0.0.1:8765`. If that port is busy it picks another and prints the address. If it cannot open a browser it writes the address into a file `winder-open-me.txt`; open that address yourself. To quit, close the program's window (or press Ctrl+C in the terminal).

10.4 Chromebook

1. Open **Settings** > **Advanced** > **Developers**, find **Linux development environment** and click **Turn on**.
2. In the same place, open **Manage USB devices** and switch the serial adapter on for Linux.
3. In the Linux terminal, type `uname -m`. If it says `x86_64` use the Linux x86_64 file, if it says `aarch64` use the aarch64 file.
4. Unpack and start it as for Linux above. You may need to add your user to the `dialout` group: `sudo usermod -aG dialout $USER`, then restart Linux.
5. Open `localhost:8765` in the Chrome browser.

NOTE

The Chromebook steps have not yet been tested on a real Chromebook.

10.5 Connection check (doctor)

If anything does not work, open the **Advanced** tab and look at **Connection check**, or run `./winder doctor` in a terminal. It tests, in order:

1. **USB serial adapter found**
2. **Allowed to use the adapter**
3. **Winder answers**
4. **Firmware is compatible**
5. **Saved settings are valid**
6. **A web browser can be opened**

Each failed line says what to do. Some have a **Fix it** button. Click **Check again** after fixing something.

10.6 The screens of the program

A **Stop motor** button is always at the top of the page. It stops anything the program started (calibration, Test mode runs). It cannot stop a wind started from the panel; use **Button 1** or

Button 2 for that.

Home
Machine status, whether the motor is on, the length wound, and the **Calibrate the machine** button.

Calibrate
The step-by-step calibration (Chapter 11).

Settings
The five editable settings (Chapter 9), with **Save to the machine**, **Undo all** and **Reset to factory settings**.

Advanced
For technicians: the **Test mode** switch, **Move the machine by hand** (**Motor on/off**, **Set length wound to zero**, **Run** a set length), **Last test winds** (**Start test winds**, **Clear table**, **Apply suggestions**), **Machine state**, **Connection check** and **What happened** (a log of recent events).

10.7 Test mode

Calibration and the **Advanced** controls put the machine in **Test mode**. The screen shows `DEBUG: HOST CONTROL` and `panel buttons locked`; **LED 2** and **LED 4** light. In Test mode:

- The panel cannot start a wind.
- **Button 1** or **Button 2** on the panel stops the motor and ends Test mode.
- If the PC stops talking to the machine for 5 seconds (cable pulled, program crashed), the machine stops the motor and leaves Test mode by itself.
- If you close the browser page, the program stops the motor and ends Test mode after about 12 seconds.

10.8 Command line options

For technicians. Type `winder -help` for the full list.

Option	Meaning
`-port`	Use this serial port instead of searching.
`-baud`	Serial speed (the machine uses 115200).
`-http-port`	Web page port (default 8765).
`-no-browser`	Do not open a browser.
`-list-ports`	List serial ports and exit.
`-fake`	Pretend machine, for practice without hardware.
`-true-coast`	With –fake: the coast, in counts, the pretend machine has.
`doctor`	Run the connection check in the terminal.
`calibrate`	Calibrate from the terminal (`-lengths`, `-measured`, `-save`).

Chapter 11

Calibration

Calibration teaches the machine two things: how far the roll *coasts* after the motor stops, and (optionally) how long an inch really is on your measuring wheel. It takes about 10 to 15 minutes.

Calibrate when: the machine is new; you have updated the firmware from a version older than fw=6; the wheel has been replaced, cleaned of build-up, or re-mounted; the motor, belt or roll core has changed; or rolls are often much longer than the label.

11.1 Before you start

- Connect the PC and start **winder** (Chapter 10). The **Home** tab should show the machine.
- Load paper exactly as for a real wind. The machine will run several short test winds and use some paper.
- Have a tape measure or ruler and a pencil ready for the length check.
- Want to see the steps first? Click **Practice (nothing moves)**.

11.2 The four steps

On **Home** click **Calibrate the machine**, or open the **Calibrate** tab.

Step 1 **Get ready.** Read the checklist on screen and confirm.

Step 2 **Test winds.** Click **Start automatic test**. The machine runs at least four short test winds, measures how far each one coasts and works out the average. It then runs a check

wind; the check passes if it stops within 20 counts (about 0.1 inch) of where it was told to. Stay clear of the machine; it starts and stops by itself. Click **Stop motor** at the top of the page if anything goes wrong.

Step 3 Check the length (optional, recommended). This corrects the wheel size.

a. Draw a pencil line across the paper at the edge of the table.

b. Click **Run**. The machine winds about 36 inches.

c. Measure from the table edge to the pencil line with a ruler, or count sheets laid along it (US letter is 11 inches long, A4 is 11.69 inches).

d. Type in what you measured.

Very small differences (under 0.2%) are ignored. A difference over 10% is refused as a measuring mistake; measure again. If the wheel size changes, the program runs the automatic test again with the new size.

Step 4 Save. Look over the new numbers and click **Save to the machine**. Only the numbers you reviewed are stored. Then click **Back to Home**.

If the program shows **Something went wrong**, read the message, then click **Try again** or **Leave**. Nothing is saved until Step 4.

11.3 What calibration changes

- **Coast after the motor stops** (`coast`), from the test winds.
- **Measuring wheel size** (`wheel_um`), from the length check if you did it.
- It never changes `margin` or `bump`, or the target lengths.

11.4 After calibrating

Wind one 30 foot roll and look at the result screen. Normally you see `** PULL SLACK **` with about 2 inches to pull, or `DONE` with a few inches over; both are normal. As an extra check, measure a roll by hand now and then and keep a note in the maintenance log (Chapter 13).

11.5 From the command line

A technician can also run `winder calibrate` in a terminal, with `-lengths`, `-measured` and `-save` (see `winder -help`).

Chapter 12

Loading new firmware (flashing)

The program inside the controller is called the *firmware*. This manual is for version fw=6. You only need to update it if your supplier sends a new version. Updating is a technician job; it needs a computer with the source repository and its tools set up (see `docs/runbooks/flash.md` in the repository).

WARNING

Switch the motor power off, or at least make sure nobody is near the roll, while you flash. The relay is opened while the controller restarts.

12.1 Steps

1. Connect the computer to the controller's **own USB port** (not the serial adapter).
2. In the repository folder run:

   `uv run python scripts/flash.py`
3. The script builds the firmware, then prints

   `Double-tap RST on the Pro Micro now.`

   Quickly press the controller board's reset button twice. The board then waits about 8 seconds for the new firmware.
4. The script finds the board and loads the firmware. When it reports success the controller restarts and shows the idle screen.

On Windows with WSL, a one-time setup is needed first: `uv run python scripts/flash.py -setup` (installs `usbipd` and asks for administrator permission once).

12.2 If flashing fails

Message	What to do
`AttachTimeout`	The board was not handed to WSL in time. Double-tap reset again sooner after the prompt; check the USB cable.
`PortTimeout`	The board's port did not appear. Double-tap again; try another USB cable or port (some cables carry power only).
`AvrdudeFailed`	Loading failed part way. Run the script again. The board can always be re-flashed with a double-tap.

12.3 After flashing

- If you came from a version older than fw=6, the machine starts on factory settings. Calibrate and save once (Chapter 11).
- Run the connection check (Section 10.5); it confirms the firmware is compatible.
- Wind one roll and check the result.

Chapter 13

Maintenance

The controller needs no routine service. The parts that matter for accurate rolls are the measuring wheel and the paper path. Always switch the power off first.

How often	Task
Every shift	Look at the measuring wheel: it must sit on the paper and turn freely. Wipe off paper dust with a dry cloth. Glance at the `DONE` overrun on the first roll.
Weekly	Clean the wheel's rubber with a lightly damp cloth; let it dry. Check that the wheel's spring or weight still presses it onto the paper. Check the encoder cable is plugged in and not rubbing.
Monthly	Measure one 30 foot roll by hand and write it in the log below. Check that all four buttons click and all four LEDs light (idle shows **LED 1** and **LED 3**; manual shows **LED 4**).
When rolls drift	Recalibrate (Chapter 11).
When the rubber is worn, cracked or glazed	Replace the wheel ([SPARE PART NUMBER]), then calibrate with the length check.

NOTE

Do not oil the measuring wheel or the paper path. Oil makes the wheel slip, and a slipping wheel winds long or short rolls without any warning.

Maintenance log

Date	Roll measured	Work done / notes	Initials

Chapter 14

Troubleshooting

Find the symptom in the left column. Try the actions in order. Switch the power off before touching the paper path. If nothing helps, write down what the screen shows and call [SUPPORT CONTACT].

Symptom	Likely cause	Action
Buttons and starting		
A push did nothing	The machine was not at the idle screen; it was just switched on; the push came too soon after another; or Test mode is on.	Check that the screen shows the idle screen (`[1] 30ft [2] 90ft`). Wait one second after power-up. Leave a short moment between pushes. If it shows `DEBUG: HOST CONTROL`, see below. If a push still does nothing, note which button and which screen, and report it.
A wind started by itself at power-up	Should not happen: a button already pushed in at power-up is ignored.	Switch off. Report it to [SUPPORT CONTACT].
A button stays pushed in after a wind	Normal: the buttons are latching switches.	Nothing to do. The next push works whichever way it rests.
Button 3 does nothing at idle	Normal: **Button 3** only pauses a running wind.	–
Button 4 does nothing during a wind	Normal: **Button 4** only works at idle or while paused.	Push **Button 3** to pause first.

continued on next page

Symptom	Likely cause	Action
Screen shows `DEBUG: HOST CONTROL`, panel will not start	A PC has the machine in Test mode.	Finish or close the program on the PC, or push **Button 1** or **Button 2** to end Test mode. It also ends by itself 5 seconds after the PC stops talking.
During a wind		
Paper jams, tears or bunches	Paper path problem. The machine does *not* detect this during the main run.	Push **Button 1** or **Button 2** at once, or switch off. Clear the jam with the power off. Treat the roll as short.
Motor does not stop when **Button 1**/**Button 2** pushed	Controller or relay fault.	**Switch the power off.** Do not use the machine; call [SUPPORT CONTACT].
Motor runs but the length stays at 0.0	Wheel not on the paper, wheel slipping, or encoder cable loose.	Stop. Check the wheel and the encoder cable and connector.
`ERR:n` on the top row, or `!! ENC MISS:n !`	Encoder signals the controller could not read: loose connector, damaged cable, electrical noise.	Check the encoder cable and its connector. Keep it away from motor wires. Measure the next roll by hand. If the number keeps rising, call [SUPPORT CONTACT].
Trim takes a long time with many bumps	Normal with the factory 6 ft coast when the real coast is much shorter (up to about 25 s); or the roll is heavy and slow.	Recalibrate (Chapter 11) to shorten the trim.
Results		
`** PULL SLACK **` and two long beeps	Normal: the wind stops about 2 inches short on purpose.	Pull the amount shown out by hand, then cut and remove the roll. It is not a fault and needs no reset.
Buttons start 30 yd / 90 yd, or no `PULL SLACK`, after an update	The machine still has settings saved by older firmware.	With the PC program click **Reset to factory settings**, then **Save to the machine** (Section 9.3).
`!! UNDER TARGET !!` with `SHORT`	Paper stopped moving during the trim (ran out, jam, wheel lifted), more than 24 bumps were needed, or the trim took over 45 seconds.	Push a button to silence. Check paper, jam, wheel. Do not sell the roll as full length (Section 5.8). If it happens on good paper, recalibrate.

continued on next page

Symptom	Likely cause	Action
`!! UNDER TARGET !!` with `STOPPED`	You (or someone) pushed **Button 1** or **Button 2** before the end.	Normal. The roll is short by the amount shown.
Rolls are much longer than the label (many inches; a few inches over or about 2 inches short is normal)	Coast setting too small, or wheel size wrong.	Recalibrate with the length check.
Rolls measure short by hand, but screen said `DONE`	Wheel slipping, dirty or worn; wheel size wrong.	Clean the wheel, check it presses on the paper, recalibrate with the length check. Replace a worn wheel.
Result vanished from the screen	It clears itself 60 seconds after a good wind (`DONE` or `PULL SLACK`). A power cut or reset also clears it.	Normal for `DONE`. After a power cut, measure the roll by hand.
Screen, LEDs and sound		
Screen garbled or blank for a moment	Electrical noise from the relay.	Wait 2 seconds; it redraws itself.
Screen stays blank or garbled	Screen cable loose, or screen fault.	Switch off and on. Check the four-wire screen cable.
Screen blank and machine restarted by itself	The controller reset itself (watchdog) after a fault. The motor is stopped and the result is lost.	Measure the roll by hand. If it repeats, call [SUPPORT CONTACT].
Fault LED lit (where fitted)	Internal controller fault.	Switch off and on. If it comes back, call [SUPPORT CONTACT].
No beep at the end	You stopped the wind yourself (stops are silent), or buzzer fault.	Check the screen. If a full wind finished with `DONE` or `PULL SLACK` and no beep, report the buzzer.
PC program		
Program says no adapter, or cannot connect	Adapter not plugged in, missing driver, wrong wiring, no permission.	Run the **Connection check** (Section 10.5) and follow its advice. Check TX to D0, RX to D1, GND to GND.

continued on next page

Symptom	Likely cause	Action
Firmware is compatible fails	The firmware is too old or too new for this program.	Update the firmware (Chapter 12) or download the latest program.
Saved settings are valid fails	Settings damaged, or firmware recently updated from before fw=6.	Calibrate and save once (Section 9.3).
Browser does not open	No default browser, or a locked-down computer.	Open `http://127.0.0.1:8765` yourself, or read `winder-open-me.txt`.
Stop motor does not stop a wind	The wind was started from the panel.	Push **Button 1** or **Button 2** on the panel.

Chapter 15

Known limits

This machine is honest about what it cannot do. Please read these points.

1. **No jam detection during the main run.** The machine only notices that the paper has stopped during the trim at the end. A jam, tear or empty supply in the middle of a wind is not detected; the motor keeps running. Stay with the machine while it winds.

2. **The guarantee trusts the measuring wheel.** If the wheel slips, is dirty, worn or lifted, the count is wrong and `DONE` can be wrong too. Keep the wheel clean and calibrated.

3. **Overrun figures are design estimates.** The expected overrun (under about 2 inches past the goal while the real coast is under about 5 feet, at most about 3 inches for a roll finished by bumps) comes from the design calculations and has not yet been confirmed on the bench. If the real coast is longer than the 6 foot setting, the roll is over by about the difference (for a 9 foot coast, about 3 feet). The trim time (up to about 25 seconds) is also an estimate.

4. **On/off motor only.** There is no speed control, so the motor always winds at full speed and coasts at the end.

5. **Screen glitches.** Relay noise can garble the screen briefly. It repairs itself within about 2 seconds; the wind is not affected.

6. **A reset loses the result.** After a power cut or watchdog reset the machine forgets the wind in progress and its result.

7. **Short by design.** The wind stops about 2 inches short of the label and asks you to pull the slack out by hand (Section 5.4). The machine cannot pull it for you.

8. **Two lengths on the panel.** Only the short (30 ft) and long (90 ft) targets are on buttons. Other lengths need the settings changed from the PC, or a manual run.

9. **Margin and legacy mode are not in the PC program.** They can only be changed by a technician over the serial command line.

10. **Chromebook use is untested.**

11. **Bench tests not yet signed off.** The expected values in the bench self-test (Appendix B) come from the design and source code, not yet from a measured machine.

Chapter 16

Specifications

Item	Value
Winding	
Preset lengths	30 ft (**Button 1**), 90 ft (**Button 2**); changeable from the PC
Safety margin	0 added to every roll (factory); wind stops 2.0 in short by design
Coast allowance	96 in = 8 ft (factory; set by calibration)
Length resolution	about 0.005 in (191 counts per inch)
Trim	bumps until the length is reached; gives up after 2 stalls, 24 bumps or 45 s
Result screen	clears after 60 s (good result); held until a push (short result)
Measuring	
Measuring wheel	4.0462 in diameter, about 12.71 in per turn
Encoder	600 lines per turn, read 4 times each = 2400 counts per turn; either direction of rotation works
Controller	
Board	Pro Micro (ATmega32U4), firmware fw=6
Screen	20 characters x 4 lines
Panel	4 latching push buttons, 4 LEDs, buzzer

Item	Value
Button debounce	about 16 ms
Motor switching	relay, on/off
Watchdog	resets the controller after 0.5 s without response; relay opens
Settings memory	kept without power, with a check code (format WIN7)
PC connection	
Serial link	115200 baud, 8 data bits, no parity, 1 stop bit; 5 V logic; wires TX, RX, GND to D0/D1
PC program	winder for Windows, macOS (Intel and Apple), Linux (x86_64 and ARM64); Chromebook via Linux (untested)
Web page	http://127.0.0.1:8765 on the PC itself
Test mode safety	motor stops if the PC is silent for 5 s
Electrical and physical	
Supply voltage and current	[POWER RATING – SEE MACHINE LABEL]
Motor	[MOTOR RATING]
Size and weight	[DIMENSIONS] / [WEIGHT]
Maximum roll diameter	[MAXIMUM ROLL DIAMETER]
Paper width	[PAPER WIDTH RANGE]

Chapter 17

Glossary

Bump
A short burst of the motor during the trim, to add the last few inches.

Calibration
Teaching the machine its coast and wheel size with a few test winds (Chapter 11).

Coast
How far the roll keeps turning after the motor is switched off.

Count (encoder count)
The machine's unit of length: one small step of the measuring wheel, about 0.005 inch.

DONE
The result screen for a complete roll that reached the label length.

Encoder
The rotation sensor on the measuring wheel.

ENC MISS / ERR
Encoder signals the controller could not read.

Firmware
The program inside the controller board. This manual covers fw=6.

Flashing
Loading new firmware into the controller.

Goal
Target length plus margin. A roll must reach the goal to be `DONE`.

Idle
Ready and waiting; the screen shows `[1] 30ft [2] 90ft`.

Latching button
A push button that stays in after one push and pops out after the next. On this machine every push counts as one click.

Legacy mode
Setting `bump` at 0: no trim and no guarantee.

Manual mode
Motor on by hand with **Button 4**, without counting.

Margin
The extra length (factory 0) added to every roll. The 2 inch slack is taken off the goal. Margin 0 means the wind aims 2 inches short of the label.

Measuring wheel
The rubber wheel riding on the paper that measures length.

Relay
The electrical switch the controller uses to turn the motor on and off.

Serial adapter
A USB gadget that lets a PC talk to the controller over three wires.

Settling
The moment after the motor stops while the roll comes to rest.

Target
The length on the label: 30 ft or 90 ft.

Test mode
The PC program has control of the motor; the panel cannot start a wind (shows `DEBUG: HOST CONTROL`).

Trim
The end of a wind: wait, check the length, bump if needed.

UNDER TARGET
The result screen for a roll that may be short.

Watchdog
A safety timer that restarts the controller if it freezes.

winder
The PC program for calibration and settings.

Appendix A

Quick-reference card

The card is on the next page, alone, so you can cut it out without losing any other page. Cut along the dashed line and post it beside the panel. Print this page single-sided.

PAPER WINDER – QUICK REFERENCE

fw=6

One push = one action. Buttons latch in and out; every push counts. Rest position does not matter. Wait 1 s after power-up.

	Idle screen	During a wind
Button 1	Start **30 ft**	**STOP**
Button 2	Start **90 ft**	**STOP**
Button 3	–	**Pause / resume**
Button 4	Manual motor on / off	When paused: nudge on / off

You see / hear	Meaning
LED 1 + **LED 3**	Idle, ready
LEDs fill 1 to 4	Winding (quarters)
LED 4 only	Manual: motor ON
LED 2 + **LED 4**, `DEBUG: HOST CONTROL`	PC Test mode; **Button 1**/**Button 2** stops
`TRIMMING...`	Final bumps, normal
2 long beeps, `** PULL SLACK **`	**Good roll, about 2 in short on purpose.** Pull the slack out by hand (screen says how much), then cut.
3 short beeps, `DONE`	**Good roll** (full length reached)
3 long beeps, repeating, `!! UNDER TARGET !! SHORT`	**Roll is SHORT.** Push any button to silence. Check paper, jam, wheel.
No beep, `STOPPED`	You stopped it early; roll is short
`ENC MISS` / `ERR`	Check encoder cable; measure next roll

EMERGENCY: switch the machine's POWER OFF. Power off before touching the paper path. The machine does not detect jams during the main run – stay with it.

Support: [SUPPORT CONTACT] Rev. 2026-10-21

Appendix B

Bench self-test summary

For technicians. The full procedures are in `docs/runbooks/hardware-tests.md` in the source repository. Each test loads one test image with `scripts/flash.py -target <name>` and checks the panel. Record pass/fail and notes for each. The expected values were worked out from the design and source code and have not yet all been confirmed on a real machine.

Test	What	Looks for	Pass?
P1	Firmware acceptance	The normal firmware: idle screen, 30/90 ft winds, pause, manual, result screens.	_____
P2	Encoder	One wheel turn changes the count by exactly 2400, both directions; no missed counts at full speed.	_____
P3	Buzzer	Beeps of 200, 500 and 1000 ms on **Button 1**, **Button 2**, **Button 3**.	_____
P4	Screen	Row order, all characters, corners, backlight.	_____
P5	LEDs and raw pins	LED chase, button pin levels.	_____
P6	Timer and buttons	Each button lights its LED; presses under 16 ms ignored; contact bounce of each latching button measured.	_____
P7	PC serial link	The machine answers the PC program's commands.	_____
P8	Watchdog and stuck screen bus	Controller recovers safely, relay open.	_____
P9	Result screen and roll speed	`settling...` then `DONE`; roll speed figure recorded.	_____
P10	Panel survives the relay	Screen repairs itself after relay noise; buttons keep working.	_____

Test	What	Looks for	Pass?
P11	Trim mode and guarantee	`TRIMMING...`, bumps, `DONE` at or past the goal; `UNDER TARGET` and alarm when the paper is held.	_____

Tested by: _____ Date: _____

Appendix C

Change history

Firmware

Version	What changed for the owner
fw=4	Result screen shows the final length after the roll stops, with a speed figure for technicians.
fw=5	Saving settings no longer pauses the machine; more diagnostics for the PC program.
fw=6	Trim mode with the minimum-amount guarantee: safety margin, final bumps, `DONE` / `!! UNDER TARGET !!` screens and the alarm. New settings format (WIN6, WIN7 from the first bench calibration): settings from older firmware are not used; calibrate once after updating. Panel buttons: every push of a latching button is one click; a button pushed in at power-up is ignored. Later fw=6 builds: factory coast 6 feet (72 in) and bumps sized from the glide measured in the same roll (shorter final bumps, trim up to 45 s); saved settings keep their old coast until **Reset to factory settings** and Save.

This manual

Revision	Changes
2026-10-20	First edition, for firmware fw=6.
2026-10-21	Factory coast 6 feet, learned bump sizing, trim limits 24 bumps and 45 s.
2026-10-21	Buttons are 30 ft and 90 ft (were 30 yd and 90 yd); lengths in feet; factory margin 0; the wind stops about 2 inches short on purpose and shows ** PULL SLACK ** with two long beeps. Boards with saved settings need **Reset to factory settings** and Save.

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